

API Usage & Integration Guide

LOGISTICS OBJECT DETECTION & REASONING REST API

1. Base URLs & Swagger Documentation

- **Live Cloudflare Tunnel Base URL:** <https://prices-debug-match-twist.trycloudflare.com/>
- **Interactive Swagger UI Console:** <https://prices-debug-match-twist.trycloudflare.com/docs>
- **Local Server Fallback:** <http://127.0.0.1:8000>

2. Endpoint Summary Table

Endpoint	HTTP Method	Content-Type	Description
/detect	POST	multipart/form-data	Returns bounding boxes, class names, and confidence scores.
/ask	POST	multipart/form-data	Natural language question answering with confidence guardrail.
/health	GET	application/json	Liveness probe returning server status.
/classes	GET	application/json	Returns the 5 supported industrial object classes.

3. Detection Endpoint (POST /detect)

cURL Example:
curl -X POST "https://prices-debug-match-twist.trycloudflare.com/detect" \ -H "Accept: application/json" \ -F "file=@warehouse_photo.jpg"

Sample Response Payload (200 OK):
{ "objects": [{ "class": "freight container", "confidence": 0.884, "bbox": { "x1": 83.64, "y1": 22.29, "x2": 587.44, "y2": 601.26 } }], "num_detections": 1, "image_size": [640, 640] }

4. Reasoning & Question Endpoint (POST /ask)

cURL Example:
curl -X POST "https://prices-debug-match-twist.trycloudflare.com/ask" \ -H "Accept: application/json" \ -F "file=@warehouse_photo.jpg" \ -F "question=How many freight containers are in this image?"

Sample Response Payload (High Confidence Answer):
{ "question": "How many freight containers are in this image?", "intent": "COUNT", "answer": "There are 2 freight containers visible.", "confidence": "high", "used_detector": true }

5. Python Integration Example

```
import requests
url = 'https://prices-debug-match-twist.trycloudflare.com/ask'
files = {'file': open('warehouse.jpg', 'rb')}
data = {'question': 'Is there a forklift near the container?'}
response = requests.post(url, files=files, data=data)
print(response.json())
```